# Seminar 1

1. Dați ex de funcție injectivă, nesurjectivă

$f: (-\infty, 7] \to [\frac{13}{2}, \infty)$

4. Din Lucrare

$A = \{10, \dots, 24\}$

$A = \bigcup_{i=1}^{5} A_i \quad A_i \cap A_j \neq \emptyset, \forall i \neq j$

$|A_i| \neq |A_j| \ (\forall) \ i \neq j$

$5 = |A| = \sum_{i=1}^{5} |A_i| = 1+2+3+4+5$

$A_1 = \{10\}$

$A_2 = \{11, 12\}$

$A_3 = \{13, 14, 15\}$

$A_4 = \{16 \dots 20\}$

$A_5 = \{21, \dots 24\}$

Lucrare

ciclu de lungime 3

$\left(\begin{smallmatrix}1&2&3\end{smallmatrix}\right)$ în $S_5$

$\begin{pmatrix}1&2&3&4&5\\3&1&2&4&5\end{pmatrix}$

$\tau = \begin{pmatrix}1&2&3&4&5&6&7&8&9&10&11\\4&9&2&1&11&6&8&3&7&5&10\end{pmatrix}$

$= (1\,4)(2\,9\,7\,8\,3)(5\,11\,10)(6)$

$\tau = (i_1, i_2 \dots i_m) \in S_n$

ciclu de lungime $m$

$\tau(i_1) = i_2$

$\tau(i_2) = i_3$

$\vdots$

$\tau(i_{m-1}) = i_m$

$\tau(i_m) = i_1$

$\tau(j) = j, \ \forall j \in \{1, \dots n\} \setminus \{i_1, \dots i_m\}$

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The Teorema $\forall \tau \in S_n \; \exists ! \; \tau = z_1 \cdots z_k$ (Tau)

cicli disjuncți de lungimi $l_i \geq 2$

$l_1 + \dots + l_k \leq n$

(!) - se scrie în mod unic

(i j) transpoziție = ciclu de lungime 2

$(1\,4)(2\,9)(9\,7)(7\,8)(8\,3)(5\,11)(11\,10)$

$(1\,2\,3) = (1\,2)(2\,3) = (2\,3)(1\,3)$

$(2\,3)(1\,3) = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 1 \end{pmatrix}\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$

$(1\,2\,3) = (2\,3\,1) = (3\,1\,2)$

$(3\,1\,2) = (3\,1)(1\,2)$

$\quad\quad\quad

$l_1 + \dots + l_k \leq 8 \quad (2)$

Din (1) și (2) $\Rightarrow$ Avem ... cazuri:

① $k = 1$
$(\tau = z_1)$, $l_1 = 4$

② $k = 2$, $l_2 = 2$
$l_1 = 4$

③ $k = 2$, $l_1 = 4$
$l_2 = 4$

④ $k = 3$, $l_1 = 4$
$l_2 = 2$
$l_3 = 2$

1. $\frac{A_8^4}{4} = \frac{\frac{8!}{(8-4)!}}{4} = X$

2. $\tau = z_1 \cdot z_2$

$\frac{A_8^4}{4} \cdot 6 = \frac{A_4^2}{2}$

3. $\frac{A_8^4}{4} \cdot \frac{A_8^4}{4} = \frac{A_8^4}{4} \cdot 6$

4. $\frac{A_8^4}{4} \cdot \frac{A_4^2}{2} \cdot 1$

Ec 0.9.

$S_3 \ni \tau = \varepsilon_1 \cdot \dots \cdot \varepsilon_k$
$e \quad l_1 \geq 2 \quad l_k \geq 2$
$l_1 + \dots + l_k \leq 3$

$\Rightarrow \tau = \varepsilon_1$ ($\varepsilon_1$ transpoziția)

$(1\,2), (1\,3), (2\,3)$

Ex 1.9. $\tau = (1\,2) \in S_6$

$z^2 = \tau$

$\operatorname{sgn}(z^2) = \operatorname{sgn}(z)^2 = 1$
$\operatorname{sgn}(\tau) = \operatorname{sgn}((1\,2)) = -1$
$\tau = (1\,2)$

$\Rightarrow$ Ec $z^2 = \tau$ nu are soluții

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$z^3 = \tau$

$(1\,2)^3 = (1\,2)$

$[(1\,2)(3\,4\,5)]^3 = (1\,2)$

Pp că $z^3 = \tau$ are soluții în $S_6$

Fie $z = z_1 \cdots z_k$ ⇒ descr în produs de cicli disjuncți

$\#$ ↓ ↓ ↓

$\ell_1 \geq 2 \quad \ell_k \geq 2 \quad \ell_1 + \cdots \ell_k \leq 6$

$z^3 = z_1^3 \cdot \ldots \cdot z_k^3 = (1\,2)$

1 ciclu $\ell = 3$

⊙ transpoziție

Transpoziție fixă $(1\,2)$

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## Seminar 2

Exerciții lucrare

L.C.R. $(\mathbb{Z}_n, +, \cdot)$ ⇒ inel com., 1.

1. $\mathbb{Z}_n$ - domeniu de integritate ⇔ $n$ e prim (⇔ $\mathbb{Z}_n$ e corp)

$U(\mathbb{Z}_n)$ ← el. inversabile din $\mathbb{Z}_n$

$n \geq 2, U(\mathbb{Z}_n) = \{\hat{x} \mid 1 \leq x \leq n, (x, n) = 1\}$

$|U(\mathbb{Z}_n)| = \varphi(n)$

↳ nr. de elemente din $U(\mathbb{Z}_n)$

$\varphi : \mathbb{N}^* \to \mathbb{N}^*$

↳ funcția indicatorul lui Euler

$\varphi(1) = 1 \; ; \; \varphi(2) = 1, \; \varphi(p) = p - 1$

↳ prim

$n \geq 2 \quad \varphi(n) = n\left(1 - \frac{1}{p_1}\right) \cdots \left(1 - \frac{1}{p_k}\right)$

$n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$

$p_1, \ldots, p_k$ prime $\neq$ 2 câte 2

$\alpha_1, \ldots \alpha_k \geq 1, \; k \geq 1$

$n = 12 \Rightarrow 12 = 2^2 \cdot 3$

$\varphi(12) = 12\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right) = 12 \cdot \frac{1}{2} \cdot \frac{2}{3} = 4$

$n = 24 \Rightarrow 2^3 \cdot 3 \quad \varphi(24) = 24 \cdot \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right) = 24 \cdot \frac{1}{2} \cdot \frac{2}{3} = 8$

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$\mathbb{Z}_m = U(\mathbb{Z}_m) \cup Z(\mathbb{Z}_m)$

( multimea divizorilor lui 0 ai lui $\mathbb{Z}_m$

$U(\mathbb{Z}_m) \cap Z(\mathbb{Z}_m) = \emptyset$

$Z(\mathbb{Z}_m) = \{\hat{x} \mid 1 \le x \le m,\ (x,m) \neq 1\}$

$|Z(\mathbb{Z}_m)| = m - \varphi(m)$

$N(\mathbb{Z}_m) = \{\hat{k} \mid (\exists) t > 0 \text{ a.i. } \hat{k}^t = \hat{0},\ k \in \mathbb{Z}_m\}$

$\hat{}$ multimea elementelor nilpotente

$\downarrow$

$\hat{k}^t = \hat{0}$

$\cup\!\!\!\cup$

$m : k^t$

$p_1^{\alpha_1} \cdots p_r^{\alpha_r} = m : k^t \Rightarrow p_1, \cdots, p_r : k^t \Rightarrow p_1 \cdots p_r : k \Rightarrow$

$\Rightarrow p_1 \cdots p_r : k$

$N(\mathbb{Z}_m) \overset{\text{sau}}{=} \{\hat{k} \mid p_1 \cdots p_r : k\}$

$\overset{\text{sau}}{\supseteq}$ $\mathbb{Z}_m$

$k = p_1 \cdots p_r \cdot x$

$t = \max_{i=1} \alpha_i \ge \alpha_i$

$k^t = p_1^t \cdots p_r^t \cdot x^t : p_1^{\alpha_1} \cdots p_r^{\alpha_r} = m \Rightarrow$

$\Rightarrow \hat{k}^t = \hat{0} \Rightarrow \hat{k} \in N(\mathbb{Z}_m)$

$m = 45 = 3^2 \cdot 5$

| 45 | 5 |
|---|---|
| 9 | 3 |
| 3 | 3 |
| 1 | |

$|U(\mathbb{Z}_m)| = \varphi(45) = 45\left(1-\frac{1}{3}\right)\left(1-\frac{1}{5}\right) = 24$

$U(\mathbb{Z}_m) = \{\hat{k} \mid 1 \le k < 45,\ (k,45) 1\}$

$|Z(\mathbb{Z}_{45})| = 45 - 24 = 21$

$Z(\mathbb{Z}_{45}) = \{\hat{0}, \hat{3}, \hat{5}, \hat{6}, \hat{9}, \hat{10}, \hat{12}, \hat{15}, \hat{20}, \hat{18}, \hat{21}, \hat{24}, \hat{25}, \hat{30}, \hat{27},$
$\hat{33}, \hat{36}, \hat{39}, \hat{35}, \hat{40}, \hat{42}\}$

$N(\mathbb{Z}_{45}) = \{\hat{0}, \hat{15}, \hat{30}\} \Rightarrow 3 \cdot 5 \cdot \mathbb{Z}_{45}$

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1. $\mathbb{Z}_n$ - domeniu de integritate $\Leftrightarrow$ $n$ e prim ($\Leftrightarrow$ $\mathbb{Z}_n$ e corp)

"$\Rightarrow$" $Z(\mathbb{Z}_n) = \{\hat{0}\} \Leftrightarrow U(\mathbb{Z}_n) = \mathbb{Z}_n \setminus \{\hat{0}\} \Rightarrow n$ e prim

$\mathbb{Z}_n$ - domeniu $\rightarrow$ ne rezultă

Cum calculăm inversul unui element din $U(\mathbb{Z}_n)$?

$n = 18 \quad U(\mathbb{Z}_{18}) = \{\hat{1}, \hat{5}, \hat{7}, \hat{11}, \hat{13}, \hat{17}\}$

$\varphi(18) = 18\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right) = 6$

$18 = 2 \cdot 3^2$

$\hat{7}^{-1} = ? \Rightarrow \hat{7} \cdot \hat{13} = \hat{91} = \hat{1}$

$7 \cdot 13 = 91 \equiv 1 \pmod{18}$

$\hat{7}^{-1} = \hat{13}$

! Calculați inversul lui $\hat{237}$ în $\mathbb{Z}_{1000}$.

2. Fie $n, a, b \in \mathbb{Z}$, $\exists\, k, l \in \mathbb{Z}$ a.î. $a \cdot k + b \cdot l = n \Leftrightarrow (a, b) \mid n$

$\Downarrow$

3. Fie $a, b \in \mathbb{Z}$, $\exists\, k, l \in \mathbb{Z}$ a.î. $a \cdot k + b \cdot l = 1 \Leftrightarrow (a, b) = 1$

$\Downarrow$

4. Fie $d = (a, b) \Rightarrow \exists\, k, l \in \mathbb{Z}$ a.î. $d = a \cdot k + b \cdot l$ ($k$ și $l$ se determină folosind algoritmul lui Euclid).

$1 = a \cdot k + b \cdot l$ (Trec la clase modulo $b$)

$a \cdot k \equiv 1 \pmod{b}$

$\hat{a} \cdot \hat{k} = \hat{1}$ în $\mathbb{Z}_b$

(am calculat inversul lui $a$ în $\mathbb{Z}_b$)

! $(237, 1000) = 1 \overset{4}{\Rightarrow} (\exists)\, k, l \in \mathbb{Z}$ a.î. $237 \cdot k + 1000 l = 1$

$\hat{237} \cdot \hat{k} = \hat{1}$

$1000 = ?$

$57 = 32 \cdot 1 + 25 \Rightarrow 25 = 57 - 32$

$32 = 25 \cdot 1 + 7 \Rightarrow 7 = 32 - 25 = 2 \cdot 32 - 57$

$25 = 7 \cdot 3 + 4 \Rightarrow 4 = 57 - 32 - (2 \cdot 32 - 57)$

$4 = 4 \cdot 57 - 7 \cdot 32$

$7 = 4 \cdot 1 + 3$

$4 = 3 \cdot 1 + \boxed{1} = (32, 57)$

$3 = 1 \cdot 3 + 0$

$3 = 2 \cdot 32 - 57 - (4 \cdot 57 - 7 \cdot 32)$

$3 = 9 \cdot 32 - 5 \cdot 57$

$1 = 4 - 3 = 4 \cdot 57 - 7 \cdot 32 - (9 \cdot 32 - 5 \cdot 57)$

$1 = 9 \cdot 57 - 16 \cdot 32$

$\boxed{1 = 57 \cdot 9 + 32 \cdot (-16)}$

$a = b \cdot q_1 + r_1 \quad 0 < r_1 < b$

$b = r_1 q_2 + r_2 \quad 0 < r_2 < r_1$

$r_{m-2} = r_{m-1} \cdot q_m + \boxed{r_m}$

$r_{m-1} = r_m \cdot q_{m+1} + 0$ $\rightarrow (a, b)$

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$1000 = 237 \cdot 4 + 52$

$237 = 52 \cdot 4 + 29$

~~$29 =$~~

$52 = 29 \cdot 2 + 23$

$29 = 23 \cdot 1 + 6$

$23 = 6 \cdot 3 + 5$

$6 = 5 \cdot 1 + \boxed{1}$

$5 = 1 \cdot 5 + 0$

$(1000, 237) = 1$

$52 \cdot \frac{4}{208}$

$\Rightarrow$

$52 = 1000 - 4 \cdot 237$

$29 = 237 - 4 \cdot 52$

$29 = 237 - 4 \cdot (1000 - 4 \cdot 237)$

$= 17 \cdot 237 - 4 \cdot 1000$

$23 = 52 - 29 \cdot 2$

$23 = 5 \cdot 1000 - 21 \cdot 237$

$6 = 29 - 23$

$6 = 38 \cdot 237 - 9 \cdot 1000$

$5 = 23 - 3 \cdot 6$

$5 = 32 \cdot 1000 - 135 \cdot 237$

$1 = 6 - 5$

$1 = 173 \cdot 237 - 41 \cdot 1000$

$237 \cdot 173 + 1000(-41) = 1$

$\hat{1} = \hat{173} \cdot \hat{237}$ în $\mathbb{Z}_{1000}$

$\hat{237}^{-1} = \hat{173}$

- Calculați restul împărțirii lui $a^b$ la $c$.

$a^b \pmod c$

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Mica thm. Fermat $p$ - prim, $a \in \mathbb{N}$ $a^p \equiv a \pmod p$

($\rightsquigarrow (a,p) = 1 (\Leftrightarrow p \nmid a)$ $a^{p-1} \equiv 1 \pmod p$)

$p$ prim

$(\mathbb{Z}_p^*, \cdot)$ - grup abelian

Lagrange $\Rightarrow \hat{a}^{p-1} = \hat{1} \Rightarrow p \mid a^{p-1} - 1 \rightarrow a^{p-1} \equiv 1 \pmod p$

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Teorema Euler $(a,m) = 1$, $a, m \in \mathbb{N} \Rightarrow a^{\varphi(m)} \equiv 1 \pmod m$

($\varphi(m) = |U(\mathbb{Z}_m)|$ ...)

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$12^{178} \pmod{13} = (-1)^{178} \pmod{13} = 1 \pmod{13}$

$12^{178} \pmod{27} = 0 \pmod{27}$

$12^{178} \pmod{31} \Rightarrow$

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$12^{178} \pmod{31}$

$(12, 31) = 1 \Rightarrow 12^{\varphi(31)} \equiv 1 \pmod{31}$ $\quad 12^{30} \equiv 1 \pmod{31}$

**Sol 1**

$12^{178} \equiv 12^{30 \cdot 5 + 28} (31) \equiv (12^{30})^5 \cdot 12^{28} (31) \equiv 12^{28} (31) \equiv$

$\equiv 144^{14} (31) \equiv 20^{14} (31) \equiv (-11)^{14} (31) \equiv 11^{14} (31) \equiv$

$\equiv 121^7 (31) \equiv (-3)^7 (31) \equiv -729 \pmod{31} \equiv -16^3 (31) \equiv -17 (31) \equiv 14 (31)$

$729 \mid 31$
$62 \quad 23$
$109$
$93$
$16$

**Sol 2** $\quad (12, 31) = 1$

$31 = 12 \cdot 2 + 7$
$12 = 7 \cdot 1 + 5$
$7 = 5 \cdot 1 + 2$
$5 = 2 \cdot 2 + 1$
$2 = 2 \cdot 1 + 0$

$7 = 31 - 12 \cdot 2$
$5 = 12 \cdot 3 - 31$
$2 = 7 - 5 = 2 \cdot 31 - 5 \cdot 12$
$1 = 12 \cdot 13 - 5 \cdot 31$

$\hat{12}^{-1} = \hat{13}$ în $\mathbb{Z}_{31}$

$12^{178} \equiv 12^{180 - 2} \pmod{31} \equiv 12^{180} \cdot 13^2 (31) \equiv (12^{30})^6 \cdot 169 (31) \equiv$

$\equiv 1 \cdot 14 (31) \equiv 14 (31)$

**Sol 3**

$12^{178} \pmod{31} \equiv -11^{89} (31) \equiv -11^{89} (31) \equiv -121^{44} \cdot 11 (31) \equiv$

$\equiv -(-3)^{44} \cdot 11 (31) \equiv -3^{44} \cdot 11 (31) \equiv -(3^5)^8 \cdot 3^4 \cdot 11 (31) \equiv$

$\equiv -(243)^8 \cdot 3^4 \cdot 11 (31) \equiv -(-3)^8 \cdot (-8) (31) \equiv 8 \cdot 3^8 (31) \equiv$

$\equiv 8 \cdot (3^3)^2 \cdot 27 (31) \equiv 8 \cdot 25 (31) \equiv 200 (31) \equiv 14 (31)$

LCR (Lema Chineză a Resturilor)

$(m_i, m_j) = 1 \quad \forall \; i \neq j$

$\mathbb{Z}_{m_1 \cdots m_k} \simeq \underbrace{\mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_k}}_{\text{produs direct de inele}}$

izom de inele

$(m_i, m_j) = 1 \; \forall \; i \neq j, \; a_i \cdots a_k \in \mathbb{Z}$

Sist. de congruențe $(*) \begin{cases} x \equiv a_1 \pmod{m_1} \\ \cdots \\ x \equiv a_k \pmod{m_k} \end{cases}$ are sol unică modulo $m_i \cdots m_k$

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Alg. rezolvare (*)

$N = m_1 \cdot \dots \cdot m_n \qquad N_i = \frac{N}{m_i} \quad (N_i, m_i) = 1$

Rezolvăm congruențele $N_i x_i \equiv 1 \pmod{m_i} \quad \forall \; i = \overline{1,n}$

Soluția unică mod $m_1 \cdot \dots \cdot m_n$ este $x = \sum_{i=1}^{n} a_i N_i x_i \pmod N$

ex. $\begin{cases} x \equiv 5 \pmod 7 \\ x \equiv 6 \pmod 8 \\ x \equiv 7 \pmod 9 \end{cases}$

$m_1 = 7, \; m_2 = 8, \; m_3 = 9$

$(7,8) = (8,9) = (7,9) = 1$

$a_1 = 5, \; a_2 = 6, \; a_3 = 7$

$N = m_1 \cdot m_2 \cdot m_3 = 7 \cdot 8 \cdot 9$

$N_1 = m_2 \cdot m_3 = 72$

$N_2 = 63$

$N_3 = 56$

$N_1 x_1 \equiv 1 \pmod 7$

$72 x_1 \equiv 1 \pmod 7$

$2 x_1 \equiv 1 \pmod 7 \Rightarrow x_1 \equiv 4 \pmod 7$

$N_2 x_2 \equiv 1 \pmod 8$

$63 x_2 \equiv 1 \pmod 8$

$-x_2 \equiv 1 \pmod 8$

$x_2 \equiv 7 \pmod 8$

$N_3 x_3 \equiv 1 \pmod 9$

$56 x_3 \equiv 1 \pmod 9$

$2 x_3 \equiv 1 \pmod 9$

$x_3 \equiv 5 \pmod 9$

$x = (5 \cdot 72 \cdot 4 + 6 \cdot 63 \cdot 7 + 7 \cdot 56 \cdot 5) \pmod{7 \cdot 8 \cdot 9}$ □

↓

soluția unică mod $(7 \cdot 8 \cdot 9)$ a sist (*)

ex: Calculați $57^{133} \pmod{130}$

$130 = 2 \cdot 5 \cdot 13$

$\begin{cases} x \equiv a_1 \pmod 2 \\ x \equiv a_2 \pmod 5 \\ x \equiv a_3 \pmod{13} \end{cases} \overset{\text{LCR}}{\Rightarrow} x \equiv _ \; (130)$

$57^{133} \pmod 2 \equiv 1 \pmod 2$

$57^{133} \pmod 5 \equiv 2^{133} \pmod 5 = (-1)^{66} \cdot 2 \pmod 5 \equiv 2 \pmod 5$

$57^{133} \pmod{13} \equiv 57^{12 \cdot 11 + 1} \pmod{13} \equiv 57 \pmod{13} \equiv 5 \pmod{13}$

$(57, 13) = 1, \quad 57^{12} \equiv 1 \pmod{13}$

$\Rightarrow \begin{cases} x \equiv 1 \pmod 2 \\ x = 2 \pmod 5 \\ x \equiv 5 \pmod{13} \end{cases} \overset{\text{LCR}}{\Rightarrow}$

$m_1 = 2, \; m_2 = 5, \; m_3 = 13$

$a_1 = 1, \; a_2 = 2, \; a_3 = 5$

$N = m_1 \cdot m_2 \cdot m_3 = 2 \cdot 5 \cdot 13$

$N_1 = 5 \cdot 13 = 65, \; N_2 = 2 \cdot 13 = 26$

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$N_3 = m_1 \cdot m_2 = 10$

$N_1 x_1 \equiv 1 (2)$

$65 \cdot x_1 \equiv 1 (2)$

$x_1 \equiv 1 (2)$

$N_2 \cdot x_2 \equiv 1 (5) \Rightarrow 26 x_2 \equiv 1 (5) \Rightarrow -1 x_2 \equiv 1 (5) \Rightarrow x_2 \equiv 1 (5)$

$N_3 \cdot x_3 \equiv 1 (13) \Rightarrow 10 x_2 \equiv 1 (13) \Rightarrow x_3 \equiv 4 (13)$

$x = \sum_{i=1}^{3} a_i N \cdot x_i \mod (130)$

$x \equiv 65 \cdot 1 \cdot 1 + 26 \cdot 2 \cdot 1 + 10 \cdot 3 \cdot 4 \pmod{130}$

$x \equiv 65 + 52 + 120 \pmod{130}$

$x \equiv 237 \pmod{130} \Rightarrow x \equiv 107 \pmod{130}$

$57^{133} \equiv 107 \pmod{130}$

1. Să se arate că pentru $\forall m \in \mathbb{N}^*$ există $m$ nr. nat consecutive a.î. niciunul dintre ele să nu fie putere a unui nr. $p$ prim.

$x+1, x+2, \dots, x+m$ sau $x, x+1, \dots, x+m-1$

$p_1 \cdot p_2 \mid x \quad (x \equiv 0 \pmod{p_1 p_2})$

$x+1 \equiv 0 \pmod{p_3 p_4}$

$\vdots$

$x+m-1 \equiv 0 \pmod{p_{2m-1} p_{2m}}$

$(p_{2i} p_{2i+1}, p_{2j} p_{2j+1}) = 1 \quad \forall i \neq j$

$$(1) \begin{cases} x \equiv 0 \pmod{p_1 p_2} \\ x \equiv -1 \pmod{p_3 p_4} \\ \vdots \\ x \equiv 1-m \pmod{p_{2m-1} p_{2m}} \end{cases}$$

Din TCR $\Rightarrow$ (1) are sol unică modulo $p_1 p_2 \cdots p_{2m}$ →

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$\to (3)$ $x \in \mathbb{N}$ a.i $x+1 \ldots x+m$ să nu fie niciunul putere a unui nr prim.

ex: Sist de congruențe

$$\begin{cases} x \equiv 0(2) \\ x \equiv 1(4) \end{cases}$$ nu are soluții

! Nu pot aplica LCR pt că $(2,4) \neq 1$
(nu sunt prime între ele)

T: Să se arate că $\forall m \geq 3$.

$$m^{m^{m^m}} - m^{m^m} \vdots 1989$$

.

$7^{9^9} \to x$

$7^{9^9} \pmod{37} \equiv 7^x \pmod{37}$ unde $7^{9^9} \equiv x(36)$

$(7,37) = 1 \overset{\text{Euler}}{\Rightarrow} 7^{36} \equiv 1(37)$

Pt. a calcula e suficient să determin $7^{9^9}(36)$

$36 = 4 \cdot 9$

$7^{9^9}(4) \equiv (-1)^{9^9}(4) \equiv -1(4) \equiv 3(4)$

$7^{9^9}(9) \equiv 7^3(9) \equiv (-2)^3(9) \equiv -8(9) \equiv 1(9)$

$(7,9) = 1 \overset{\text{Euler}}{\Rightarrow} 7^{\varphi(9)} \equiv 1(9) \Rightarrow 7^6 \equiv 1(9)$

$\varphi(9) = 9\left(1 - \frac{1}{3}\right) = 6$

$6 = 2 \cdot 3$

$9^9(3) \equiv 0(3)$

$9^9(2) \equiv 1(2)$

$3\ell = 2t + 1$

$t = \frac{3\ell - 1}{2}$

$\ell = 2\ell_1 + 1$

$9^9 \equiv 3(6)$

$3\ell = 6\ell_1 + 3$

$4a + 3 = 9b + 1$

$4a = 9b - 2$

$a = \frac{9b-2}{4} \in \mathbb{N}$ $b = 4b_1 + 2$

$9b \equiv 2(4)$

$b \equiv 2(4)$

din 2 în 2

$7^{9^9} \equiv 19(36) \Rightarrow 7^{19} \pmod{37}$

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Tema: $a^b(c)$

$a^{b^{cd}}(e)$

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# Seminar 3

Rez. sist de congr.

$$\begin{cases} 5x \equiv 2(7) \\ 3x \equiv 6(8) \\ 7x \equiv 3(9) \end{cases}$$

$m_1 = 7 \quad m_2 = 8 \quad m_3 = 9$

$(7,8) = (8,9) = (7,9) = 1$ (prime intre ele)

$a_1 = 2 \quad a_2 = 6 \quad a_3 = 9$

$$N_i = \frac{N}{m_i}$$

$$N_1 = \frac{m_1 \cdot m_2 \cdot m_3}{m_1} = m_2 \cdot m_3 = 8 \cdot 9 = 72$$

$$N_2 = \frac{m_1 \cdot m_2 \cdot m_3}{m_2} = 7 \cdot 9 = 63$$

$$N_3 = \frac{m_1 \cdot m_2 \cdot m_3}{m_3} = 7 \cdot 8 = 56$$

~~$N_1 \cdot x_1 \equiv$~~

$(a, m) = 1$

$ax \equiv b \pmod m$

Există $x, y \in \mathbb{Z}$, $ax + by = m \Leftrightarrow (a,b) \mid m$

$(a,b) = 1 \overset{\text{Bez}}{\underset{\text{Euclid}}{\Leftrightarrow}} (\exists) k, l \in \mathbb{Z} \quad ak + bl = 1$

$(a,b) = 1$, $ax \equiv 1(b)$, $\bar{a}\bar{k} + \bar{b}\bar{l} = \bar{1}$

$ak \equiv 1(b)$ $\quad \bar{a}\bar{k} = \bar{1}$ in $\mathbb{Z}_b$

$$\begin{cases} 5x \equiv 2(7) \\ 3x \equiv 6(8) \\ 7x \equiv 3(9) \end{cases}$$

$5x \equiv 2(7) \Rightarrow x \equiv _ \cdot 2(7)$

$(5,7) = 1 \Rightarrow 5 \cdot k + 7 \cdot l = 1$

$\hat{5} \cdot \hat{k} = \hat{1}$

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$7 = 5 \cdot 1 + 2$

$5 = 2 \cdot 2 + 1$

$2 = 1 \cdot 2 + 0$

$1 = 1 + 0$

$2 = 7 - 5$

$\Rightarrow$ ~~$5 = 7 - 2$~~

$1 = 5 - 2$

$1 = 5 - 2(7 - 5)$

$7 = 5 \cdot 1 + 2$

$5 = 2 \cdot 2 + 1$

$2 = 7 \cdot 5$

$3x \equiv 6(8) \Rightarrow (3,8) = 1 \Rightarrow 3 \cdot \hat{k} + 8 \cdot \hat{l} = \hat{1}$

$\hat{3}\hat{k} = \hat{1}$

$$\begin{cases} x \equiv \hat{3} \cdot 2(7) \\ x \equiv \hat{3} \cdot 6(8) \\ x \equiv 43(9) \end{cases} \Rightarrow \begin{cases} x \equiv 6(7) \\ x \equiv 18(8) \\ x \equiv 12(9) \end{cases} \Rightarrow \begin{cases} x \equiv 6(7) \\ x \equiv 2(8) \\ x \equiv 3(9) \end{cases}$$

$m_1 = 7, \quad m_2 = 8, \quad m_3 = 9$

$a_1 = 6 \quad a_2 = 2, \quad a_3 = 3$

$N_i = \frac{N}{m_i}$

Sistemul de congruențe $\begin{cases} x \equiv 1(2) \\ x \equiv 5(7) \\ x \equiv 6(10) \end{cases}$ nu are soluție!

$N_1 = m_2 \cdot m_3 = 72$

$N_2 = 7 \cdot 9 = 63$

$N_3 = 7 \cdot 8 = 56$

$N_1 \cdot x_1 \equiv 1(7)$

$72 \cdot x_1 \equiv 1(7)$

$2 x_1 \equiv 1(7)$

$x_1 \equiv 4(7)$

$63 x_2 \equiv 1(8)$

$-x_2 \equiv 1(8)$

$x_2 \equiv 7(8)$

$56 x_3 \equiv 1(9)$

$2 x_3 \equiv 1(9)$

$x_3 \equiv 5(9)$

$$x = \sum_{i=1}^{3} N_i a_i x_i \cdot (7 \cdot 8 \cdot 9)$$

$x = (72 \cdot 6 \cdot 4 + 63 \cdot 2 \cdot 7 + 3 \cdot 56 \cdot 5)(7 \cdot 8 \cdot 9)$

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Rezolvați în mulțimea numerelor întregi ecuația:

$$ax + by = m$$
$$a, b, m \in \mathbb{Z}$$

$ax + by = m$ $\qquad$ $3x + 5y = 36$

$ax + by = m$ are soluții $\Leftrightarrow (a, b) \mid m$

$3x + 5y = 36$

$3x_1 + 5x_2 = 36$ pentru ④

$$\begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = b_1 \\ \dots \\ a_{m1}x_1 + \dots + a_{mn}x_n = b_m \end{cases}$$

sist de ec liniare

sist e compatibil

M. sol = $\{(y_1 \dots y_n) + $ sol partic | orice soluție a sistemului omogen asociat

(*) $3x + 5y = 1$

$5 = 3 \cdot 1 + 2 \Rightarrow 2 = 5 - 3$

$3 = 2 \cdot 1 + 1 \Rightarrow 1 = 3 - 2$

$2 = 2 \cdot 1 + 0$

$1 = 3 - 5 + 3$

$1 = 2 \cdot 3 - 5$

O soluție particulară e dată de alg lui Euclid.

$3 \cdot 2 + (-1) \cdot 5 = 1$

Ec (*) are sol particulară

$$\begin{cases} x_0 = 2 \\ y_0 = -1 \end{cases}$$

Fie $x, y$ a.i. $3x + 5y = 0 \Rightarrow 3x = -5y$ $\quad (5, 3) = 1$, $5 \mid 3x \Rightarrow 5 \mid x \Rightarrow$

$\Rightarrow x = 5k$

$y = -3k$ $\quad k \in \mathbb{Z}$

Sol ec $3x + 5y = 0$ sunt $\{(5k, -3k) \mid k \in \mathbb{Z}\}$

Soluția ec (*) se obține

$\{(2, -1) + (5k, -3k) \mid k \in \mathbb{Z}\} =$

$= \{(2 + 5k, -1 - 3k) \mid k \in \mathbb{Z}\}$

$3x + 5y = 36$

$3 \cdot 2 + 5 \cdot -1 = 1 \mid \cdot 36$

$3 \cdot 2 \cdot 36 + 5 \cdot -36 = 36$

$3 \cdot 72 + 5 \cdot -36 = 36 \Rightarrow$ Soluțiile întregi ale ec $3x + 5y = 36$

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sunt $\{(+2,-3\cdot 6) + (5k,-3k) \mid k \in \mathbb{Z}\} \Rightarrow$

$\Rightarrow \{(+2-5k, -36-3k) \mid k \in \mathbb{Z}\}$.

**Tema:** Dați ~~toate~~ soluțiile

$ax+by=n$

$ax+bm=n$ are sol $\Leftrightarrow (a,b) \mid n$

Not $(a,b)=d \rightsquigarrow a=da_1, b=db_1, (a_1,b_1)=1$

$a_1x+b_1y=n_1 \qquad n=dn_1$

**Rezolvare:** Fie $x_0, y_0$ sol particulară

a ec $a_1x+b_1y=n_1$ ($y_0$ det cu alg lui Euclid)

Sol. ecuației sunt $\{(x_0,y_0)+(b_1k,-a_1k) \mid k \in \mathbb{Z}\}$.

$a_1x+b_1y=0 \Rightarrow b_1 \mid a_1x$, $(b_1,a_1)=1$ $\Rightarrow b_1 \mid x \Rightarrow x=b_1k$

$y=-a_1k$

## Operații cu ideale

1. Fie $R$ inel com 1.

$I \neq \emptyset$

$I \subseteq R$ $I$ s.n. ideal dacă $\begin{cases} 1. (I,+) \leq (R,+) \text{ (subgrup)} \\ 2. \forall a \in I, \forall x \in R \Rightarrow a \cdot x \in I \end{cases}$

Exemple:

1. $I = 2\mathbb{Z} \cup 5\mathbb{Z} = \{k \in \mathbb{Z} \mid 2 \mid k \text{ sau } 5 \mid k\}$

Subgrup $\Rightarrow \forall x,y \in I \Rightarrow x-y \in I$

$5-2=3 \notin I \Rightarrow$ $I$ nu satisface 1 $\Rightarrow$

$5 \in I$, $2 \in I$

$\Rightarrow I$ nu e ideal al lui $R$.

2. $I = 2\mathbb{Z} \cap 5\mathbb{Z} = \{k \in \mathbb{Z} \mid 2 \mid k \text{ și } 5 \mid k\}$

$12-2=10 \notin I \Rightarrow I$ nu e ideal.

$12 \in I$, $2 \in I$

3. $I = 16\mathbb{Z} \cap 24\mathbb{Z} = \{k \in \mathbb{Z} \mid 16 \mid k \text{ și } 24 \mid k\}$

$= 48\mathbb{Z}$ (arătăm)

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**Ideale lui $\mathbb{Z}$:** $n\mathbb{Z}$, $n \in \mathbb{N}$ $(-n\mathbb{Z} = n\mathbb{Z})$

$16\mathbb{Z} \cap 24\mathbb{Z} = 48\mathbb{Z}$ - ideal al lui $\mathbb{Z}$

Operații cu ideale:

$I, J \trianglelefteq R$ — ideale ale lui $R$

$I \cap J$, $J \cdot J$, $I + J$, $I : J$, $I \cup J$

$I \cap J$ ideal al lui $R$.

$J \cdot J$ ideal al lui $R$

$I + J$ ideal al lui $R$

$I : J$ ideal al lui $R$

$I \cup J$ ideal al lui $R$ $\Leftrightarrow$ (1) $I \subseteq J$ sau $J \subseteq I$

"$\Rightarrow$" Pp prin reducere la absurd că

$I \not\subseteq J$ și $J \not\subseteq I$ $\Rightarrow$

$\Rightarrow (\exists)\ x \in I \setminus J$ și $y \in J \setminus I$.

$x, y \in I \cup J \overset{(1)}{\Longrightarrow} x - y \in I \cup J$, $I \cup J \trianglelefteq R$

(1) $x - y \in I$ sau (2) $x - y \in J$

I. $x - y \in I$

$x \in I$, $I \trianglelefteq R \Rightarrow x - (x - y) \in I \Rightarrow y \in I$ ↯

↯ $(y \in J \setminus I)$

II analog.

$x - y \in J$, $y \in J$, $J \trianglelefteq R \Rightarrow (x - y) - y \in J \Rightarrow x \in J$ ↯.

Deci $I$ și $II$ $\Rightarrow$ pp e falsă $\Rightarrow I \subseteq J$ sau $J \subseteq I$.

$I \cdot J \overset{def}{=} \{ \sum_{\text{finită}} a \cdot b \mid a \in I, b \in J \} =$

$= \{ \sum_{i=1}^{n} a_i b_i \mid a_i \in I, b_i \in J, \forall i = \overline{1,n}, n \in \mathbb{N}^* \}$

$(ex: I \cdot J \trianglelefteq R)$

$I + J \overset{def}{=} \{ a + b \mid a \in I, b \in J \} \trianglelefteq R$ (Exc.)

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$I : J \overset{\text{def}}{=} \{ a \in R \mid a \cdot J \subseteq I \} \trianglelefteq R$.

Demonstratie $I : J \trianglelefteq R$. Trebuie să verificăm $\begin{cases} 1. \forall x, y \in I : J \Rightarrow \\ \Rightarrow x - y \in I : J \\ 2. \forall x \in I : J \\ \forall a \in R \Rightarrow x \cdot a \in I : J \end{cases}$

$a \cdot J = \{ a \cdot j \mid j \in J \}$

1. $\forall x, y \in I : J \Rightarrow x - y \in I : J$.

Fie $x, y \in I : J \overset{\text{def}}{\underset{I:J}{\Rightarrow}} x \cdot J \subseteq I$ și $y \cdot J \subseteq I$

Vrem să arăt $x - y \in I : J \overset{\text{def}}{\underset{I:J}{\Leftrightarrow}} (x - y) \cdot J \subseteq I \Leftrightarrow$

$(\Leftrightarrow (x - y) a \in I \ , \ \forall a \in J)$

Fie $a \in J$

$(x - y)a = x \cdot a - y \cdot a$

$x \in I : J \ , \ a \in J \Rightarrow xa \in I$

$y \in I : J \ , \ a \in J \Rightarrow ya \in I$

$\overset{1)}{\underset{I \trianglelefteq R}{\Rightarrow}} xa - ya \in I \Rightarrow (x - y)a \in I$

$\Rightarrow x - y \in I : J$.

2. Fie $x \in I : J$

$a \in R$

Vrem să arăt că $x \cdot a \in I : J \overset{\text{def}}{\underset{I:J}{\Leftrightarrow}} (x \cdot a) \cdot J \subseteq I \Leftrightarrow$

$\Leftrightarrow (xa) \cdot b \in I \ , \ \forall b \in J$.

Fie $b \in J$.

$(x \cdot a) \cdot b \overset{\text{asoc}}{\underset{1.1}{=}} x(a \cdot b) \in I$

$x \in I : J \overset{\text{def}}{\underset{I:J}{\Rightarrow}} x \cdot J \in I$

$a \in R$, $a \in J$ $\overset{2)}{\underset{J \trianglelefteq R}{\Rightarrow}} a \cdot b \in J$

$\Rightarrow x \cdot a \cdot b \in I$

Din 1 și 2 $\Rightarrow J : J$ e ideal

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$R = \mathbb{Z}$ $\quad I = m\mathbb{Z}$ , $J = m\mathbb{Z}$

$m\mathbb{Z} \cap m\mathbb{Z} \overset{?}{=} \mathbb{Z} = [m, m]\mathbb{Z}$ → c.m.m.m.c

$m\mathbb{Z} \cdot m\mathbb{Z} \overset{?}{=} \mathbb{Z} = m \cdot m\,\mathbb{Z}$

$m\mathbb{Z} + m\mathbb{Z} \overset{?}{=} \mathbb{Z} = (m, m)\mathbb{Z}$ → c.m.m.d.c

$m\mathbb{Z} : m\mathbb{Z} \overset{?}{=} \mathbb{Z} = \frac{m}{(m,m)}\mathbb{Z}$

---

$4\mathbb{Z} \cap 6\mathbb{Z} = 12\mathbb{Z}$

$4\mathbb{Z} + 6\mathbb{Z} = \{4k + 6\ell \mid k, \ell \in \mathbb{Z}\}$

$4\mathbb{Z} : 6\mathbb{Z} = \frac{4}{(4,6)}\mathbb{Z}$ . $[m, m]$ $\quad d\, m_1 m_1$

$(m, m) = d$
↳ $m = d m_1$ , $m = d m_1$
$(m_1, m_1) = 1$

---

Demonstratie:

- $m\mathbb{Z} : m\mathbb{Z} = \frac{m}{(m,m)}\mathbb{Z}$

"$\subseteq$" Fie $x \in m\mathbb{Z} : m\mathbb{Z}$

$x \cdot m\mathbb{Z} \subseteq m\mathbb{Z} \Rightarrow x \cdot m = m \cdot k$ , $k \in \mathbb{Z}$

$x \cdot m_1 d\,\mathbb{Z} \subseteq m d\,\mathbb{Z}$ $\quad x m_1 d = m_1 d k$

$x m_1 = m_1 k$

$\left.\begin{matrix} m_1 \mid x m_1 \\ (m_1, m_1) = 1 \end{matrix}\right| \Rightarrow m_1 \mid x \Rightarrow x \in m\mathbb{Z}$

"$\supseteq$" Fie $x \in m_1\mathbb{Z} \Rightarrow x = m_1 k$

Fie $\ell \in \mathbb{Z}$ $x \cdot m\ell = mk \cdot m\ell = mk \cdot m \cdot d\ell = mk\, m\ell : m \Rightarrow$

$\Leftrightarrow x \in m\mathbb{Z} \subseteq m\mathbb{Z}$

$\Leftrightarrow x \cdot m\ell \in m\mathbb{Z}, \forall \ell$

$x \in m\mathbb{Z} : m\mathbb{Z}$ ⊆

- $m\mathbb{Z} \cap m\mathbb{Z} = [m, m]\mathbb{Z}$

"$\subseteq$" Fie $x \in m\mathbb{Z} \cap m\mathbb{Z} \Rightarrow x \in m\mathbb{Z}$ și $x \in m\mathbb{Z}$

$\left.\begin{matrix} x \in m\mathbb{Z} \Rightarrow m \mid x \\ x \in m\mathbb{Z} \Rightarrow m \mid x \end{matrix}\right\} \Rightarrow [m, m] \mid x \Rightarrow$

$\Rightarrow \exists\, t \in \mathbb{Z}$ a.i. $x = [m, m] \cdot t$ , de unde

$x \in [m, m]\mathbb{Z} \Rightarrow m\mathbb{Z} \cap m\mathbb{Z} \subseteq [m, m]\mathbb{Z}$ 1

"$\supseteq$" Fie $x \in [m, m]\mathbb{Z} \Rightarrow [m, m] \mid x \Rightarrow$

$\Rightarrow \begin{cases} m \mid x \\ m \mid x \end{cases} \Rightarrow \begin{cases} x \in m\mathbb{Z} \\ x \in m\mathbb{Z} \end{cases} \Rightarrow x \in m\mathbb{Z} \cap m\mathbb{Z}$

Cum $x \in [m, m]\mathbb{Z} \Rightarrow [m, m]\mathbb{Z} \subseteq m\mathbb{Z} \cap m\mathbb{Z}$ 2.

Din 1 și 2 avem că $m\mathbb{Z} \cap m\mathbb{Z} = [m, m]\mathbb{Z}$.

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• $n\mathbb{Z} \cdot m\mathbb{Z} = n \cdot m\mathbb{Z}$

$n\mathbb{Z} \cdot m\mathbb{Z} = \{ \sum_{i=1}^{r} | a, b, a \in n\mathbb{Z}, b \in m\mathbb{Z}, \forall r \in \mathbb{N}^* \}$

Fie $x \in n\mathbb{Z} \cdot m\mathbb{Z} \Rightarrow x = \sum_{i=1}^{r} a_i \cdot b_i \Rightarrow$

$\Rightarrow x = \sum_{i=1}^{r} n \cdot c_i \cdot m \cdot d_i = \sum_{i=1}^{r} n \cdot m \cdot c_i d_i =$

$= n \cdot m \cdot \sum_{i=1}^{r} c_i d_i \in n \cdot m\mathbb{Z}$

Fie $y \in nm\mathbb{Z}$

$y = (n \cdot m)k, k \in \mathbb{Z}$

$n \cdot (mk) \in n\mathbb{Z} \cdot m\mathbb{Z}$

($n\mathbb{Z}$, $m\mathbb{Z}$)

$I \cdot J = \{ \sum_{i=1}^{r} a_i b_i | a \in I, b_i \in J \}$ $r \in \mathbb{N}^*$

$n\mathbb{Z} = \{ nk | k \in \mathbb{Z} \}$

• $n\mathbb{Z} + m\mathbb{Z} = (n, m)\mathbb{Z}$ $(= d\mathbb{Z})$

"$\geq$" Alg Euclid

$d = (n, m) \Rightarrow d = n \cdot k + m \cdot \ell$ p/ $k, \ell \in \mathbb{Z}$

Fie $dt \in d\mathbb{Z} \Rightarrow dt = n \cdot (tk) + m(t\ell)$

($n\mathbb{Z} + m\mathbb{Z}$, $n\mathbb{Z}$, $m\mathbb{Z}$)

"$\subseteq$" Fie $x \in n\mathbb{Z} + m\mathbb{Z} \Rightarrow$

$\Rightarrow x = nk' + m\ell' = dn_1k' + dm_1\ell' = d(n_1k' + m_1\ell')$

($d\mathbb{Z}$)

Calculați idealul:

1) $((10\mathbb{Z} \cap 23\mathbb{Z}) \cdot 14\mathbb{Z}) + 7\mathbb{Z}$

2) $(((4, 6, 14)\mathbb{Z} \cdot 3\mathbb{Z}) + (8\mathbb{Z} \cap 10\mathbb{Z})) : 15\mathbb{Z}) \cap 20\mathbb{Z}$

Rezolvare:

1) $((10\mathbb{Z} \cap 23\mathbb{Z}) \cdot 14\mathbb{Z}) + 7\mathbb{Z} = (230\mathbb{Z} \cdot 14\mathbb{Z}) + 7\mathbb{Z} =$

$= 230 \cdot 14\mathbb{Z} + 7\mathbb{Z} = (230 \cdot 14, 7)\mathbb{Z} = 7\mathbb{Z}.$

Temă $(((a_1\mathbb{Z} : a_2\mathbb{Z}) \cap a_3\mathbb{Z}) + (a_4\mathbb{Z} \cdot a_5\mathbb{Z})) : (a_6\mathbb{Z} + a_7\mathbb{Z})$

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2. $((((4,6,14)\mathbb{Z} \cdot 3\mathbb{Z}) + (8\mathbb{Z} \cap 10\mathbb{Z}) : 15\mathbb{Z}) \cap 20\mathbb{Z} =$

$= (2\mathbb{Z} \cdot 3\mathbb{Z}) + (8\mathbb{Z} \cap 10\mathbb{Z}) : 15\mathbb{Z}) \cap 20\mathbb{Z} =$

$= 6\mathbb{Z} + 40\mathbb{Z} : 15\mathbb{Z} \cap 20\mathbb{Z}$

$= 2\mathbb{Z} : 15\mathbb{Z} \cap 20\mathbb{Z} = 2\mathbb{Z} \cap 20\mathbb{Z} = 20\mathbb{Z}$

R inel $a_1 \dots a_m \in R$

$(a_1, \dots, a_m)$ <u>not</u> $(a_1, \dots a_m)R = \{\sum_{i=1}^{m} a_i b_i \mid b_i \in R\} =$

= idealul generat de $a_1, \dots a_m$

$a_1R + a_2R + \dots + a_mR = \{a_1b_1 + a_2b_2 + \dots + a_nb_m \mid b_1 \dots b_m \in R\}$

$(4,6,14)\mathbb{Z} = 4\mathbb{Z} + 6\mathbb{Z} + 14\mathbb{Z} =$

$= 2\mathbb{Z} + 14\mathbb{Z} = (2,14)\mathbb{Z} = 2\mathbb{Z}$

Temă:

1. $(((25\mathbb{Z} : 22\mathbb{Z}) \cap 14\mathbb{Z}) + (37\mathbb{Z} \cdot 84\mathbb{Z})) : (31\mathbb{Z} + 17\mathbb{Z}) =$

$(\frac{25}{\dots}\mathbb{Z} \dots$

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$N_2\ x \equiv 5\ (8)$

$91x \equiv 5\ (8)$

$3x \equiv 5\ (8)$

$x \equiv 7\ (8)$

$N_3\ x \equiv 1\ (7)$

$104\ x \equiv 1\ (7)$

$6x \equiv 1\ (7)$

$x \equiv 6\ (7)$

$$x = \sum_{i=1}^{m} N_i \cdot a_i \cdot x_i \quad (13 \cdot 8 \cdot 7)$$

$$x = (56 \cdot 8 \cdot 2 + 91 \cdot 5 \cdot 7 + 104 \cdot 1 \cdot 6) \quad (13 \cdot 8 \cdot 7)$$

07.03.2022

## Seminar 4

### Inel factor T.F.I

$R$ - inel com. $\quad a \in R \quad (a) \overset{not}{=} aR = \{ar \mid r \in R\}$

$I \trianglelefteq R$

$R/I = \{\hat{a} \mid a \in R\}$ $\qquad \hat{a} = \hat{b}$ în $R/I \overset{RLS}{\iff} a - b \in I$

$\Updownarrow$

$(a \equiv b \pmod I)$

$R = \mathbb{Z} \qquad I = m\mathbb{Z}$

$\hat{a} = \hat{b}$ în $\mathbb{Z}_m = \mathbb{Z}/m\mathbb{Z} \iff a - b \in m\mathbb{Z} \iff m \mid a - b$

$\Updownarrow$

$a \equiv b \pmod m$

$\mathbb{Z}_m = \{\hat{k} \mid k \in \mathbb{Z}\} = \{\hat{0}, \hat{1}, \hat{2} \dots \widehat{m-1}\}$

$\{0, 1, 2 \dots m-1\}$ este SciR pt rel de echiv $\equiv \pmod m$ (*)

SciR - sistem complet independent de reprezentanți

$(R/I, +, \cdot)$, $\quad \hat{a} + \hat{b} = \widehat{a+b}$, $\quad \hat{a} \cdot \hat{b} = \widehat{a \cdot b}$

↑ inel factor

### TFI (teorema fundamentală de izomorfism)

$\varphi : A \to S$ morfism de inele $\Rightarrow A/\ker\varphi \simeq \operatorname{Im}\varphi$

---

$R[X]$ - inelul de polin. într-o nedet.

$0 \neq f(x) \in R[X] \qquad LC(f) \in U(R)$

$I = (f(x)) = \{f(x) \cdot g(x) \mid g(x) \in R[X]\}$

$R[X]/I = \{h(x) \mid h \in R[X]\} = \{\widehat{r(x)} \mid r(x) \in R[X], \operatorname{grad}(r(x)) < \operatorname{grad} f(x)\}$ (**)

(*) Pentru $R/I$ un SciR este o submulțime $S \subseteq R$ care

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satisface:

c: 1. $\forall a \in R \ \exists s \in S \quad a \equiv s \pmod I \Leftrightarrow (\hat{a} = \hat{s})$

i: 2. $\forall s_1 + s_2 \in S \quad s_1 \not\equiv s_2 \pmod I$

$S$

$\hat{s_1} \neq \hat{s_2}$

$\overset{(**)}{\Rightarrow}$ Aratăm că $\Delta \{ r(x) \mid r(x) \in R[x] \ \text{grad}(r(x)) < \text{grad}(f(x)) \}$

$\Delta$ este un SC1R pt $R[x]/(f(x))$

---

$\hat{h_1(x)} = \hat{h_2(x)}$ în $R[x]/f(x) \overset{\text{def}}{\Leftrightarrow} h_1(x) - h_2(x) \in (f(x)) \Rightarrow$

$\Rightarrow h_1(x) - h_2(x) = f(x) \cdot q(x)$ pt un $q(x) \in R[x]$

---

(**) Tr să verific 1 și 2.

1. Fie $g(x) \in R[x] \Rightarrow$

ex în $\mathbb{Z}_m$

Fie $k \in \mathbb{Z} \quad k = m \cdot q + r \quad 0 \leq r < m$

$k - r = m \cdot q \Rightarrow \hat{k} = \hat{r}$

$\Rightarrow$ aplic teorema împărțirii cu rest pt $g$ la $f$.

$g(x) = f(x) \cdot q(x) + r(x) \quad \text{grad}(r(x)) < \text{grad}(f(x))$

$g(x) - r(x) = f(x) \cdot q(x) \in I$

$\overset{\text{def}}{\Leftrightarrow} \hat{g(x)} = \hat{r(x)}$ în $R[x]/(f(x))$

$\text{grad}(r(x)) < \text{grad}(f(x))$.

$\Rightarrow \Delta$ este un sistem complet de reprezentanți

2. Fie $g_1(x), g_2(x) \in \Delta$ a.î. $\hat{g_1(x)} = \hat{g_2(x)}$ în $R[x]/I$.

$\overset{\text{def } I}{\Rightarrow} g_1(x) - g_2(x) \in (f(x)) \Rightarrow^{(*)} g_1(x) - g_2(x) = f(x) \cdot h(x)$ pt $h(x) \in R[x]$

$\text{grad}(g_1(x) - g_2(x)) \leq \max\{\text{grad}(g_1(x)), \text{grad}(g_2(x))\} < \text{grad}(f(x))$ (3)

Dacă $h(x) \neq 0 \Rightarrow \text{grad}(f(x) \cdot h(x)) = \text{grad}(f(x)) + \text{grad}(h(x)) \geq \text{grad}(f(x))$ (4)

$\downarrow$

$LC(f) \in U(R)$

$h(x) \neq 0 \rightarrow \text{grad}(h(x)) \geq 0$

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Din (*) via (3), (4) $\Rightarrow h(x) = 0 \Rightarrow g_1(x) = g_2(x) \Rightarrow$

$\Rightarrow \Delta$ este un sistem independent de reprezentanți.

**Aplicații:**

1. $\underset{a \in R}{R[X]/(X-a)} = \{\hat{b} \mid b \in R, \text{grad } \hat{b} < \text{grad}(g(x))\}$ ($\text{grad } \hat{b} < 1$)

2. $\mathbb{Z}[X]/(X^2+1) = \{\widehat{g(x)} \mid g(x) \in \mathbb{Z}[X], \text{grad}(g(x)) < \text{grad}(X^2+1)\}$
$= \{\widehat{ax+b} \mid a, b \in \mathbb{Z}\}$

3. $\mathbb{Z}[X]/(X^3-2) = \{\widehat{ax^2+bx+c} \mid a, b, c \in \mathbb{Z}\}$

4. $\mathbb{Z}[X]/(X^2-3) = \{\widehat{ax+b} \mid a, b \in \mathbb{Z}\} \rightsquigarrow (a, b) \in \mathbb{Z} \times \mathbb{Z}$

$\widehat{ax+b} + \widehat{cx+d} = \widehat{ax+b+cx+d} = \widehat{(a+c)x+b+d}$

$\widehat{ax+b} \cdot \widehat{cx+d} = \widehat{acx^2 + x(ad+bc) + bd} \overset{(*)}{=} \widehat{x(ad+bc) + bd + 3ac}$

$$\begin{array}{l|l} acx^2 + x(ad+bc) + bd & x^2 - 3 \\ \hline -acx^2 + 3ac & ac \\ \hline / \; x(ad+bc) + bd + 3ac & \end{array}$$

$\widehat{x^3 - 3x + 1} \cdot \widehat{x^4 - x} =$

$= \hat{1} \cdot \widehat{(-x+9)} = \widehat{9-x}$

$$\begin{array}{l|l} x^3 - 3x + 1 & x^2 - 3 \\ \hline -x^3 + 3x & x \\ \hline = \; = \; 1 & \end{array}$$

(*) $\widehat{x^2 - 3} = \hat{0} \Rightarrow \hat{x}^2 = \hat{3} \mid \cdot \hat{x}$

$\hat{x}^3 = \hat{3}\hat{x}$

$$\begin{array}{l|l} x^4 - x & x^2 - 3 \\ \hline -x^4 + 3x^2 & x^2 + 3 \\ \hline / \; 3x^2 - x & \\ -3x^2 + 9 & \\ \hline -x + 9 & \end{array}$$

5. Dați o structură de inel pe mulțimea $\mathbb{Z} \times \mathbb{Z}$ a.î. inelul obținut să fie izomorf cu $\mathbb{Z}[X]/(X^2-3)$

$\varphi : \mathbb{Z}[X]/X^2-3 \to \mathbb{Z} \times \mathbb{Z}$

$\varphi(\widehat{ax+b}) = (a, b)$ $\varphi$ bijectie

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Vreau ca $\varphi$ să fie și morf. de inele

$$\varphi(\widehat{ax+b+cx+d}) = \varphi(\widehat{ax+b}) + \varphi(\widehat{cx+d})$$
$$\overset{n}{(a,b)} + \overset{n}{(c,d)}$$

și

$$\varphi(\widehat{ax+b}\cdot\widehat{cx+d}) = \varphi(\widehat{ax+b}) \cdot \varphi(\widehat{cx+d})$$
$$\overset{n}{(a,b)} \cdot \overset{n}{(c,d)}$$

Cele 2 operații pe $\mathbb{Z}\times\mathbb{Z}$ se definesc astfel:

$$(a,b)+(c,d) = \varphi(\widehat{ax+b} + \widehat{cx+d}) = \varphi(\widehat{(a+c)x+b+d}) \overset{\text{def}}{=\!=}_{\varphi} (a+c, b+d)$$

$$(a,b)\cdot(c,d) = \varphi(\widehat{ax+b}\cdot\widehat{cx+d}) = \varphi((ad+bc)x+(bd+3ac)) \overset{\text{def}}{=\!=}_{\varphi}$$
$$= (ab+bc,\ bd+3ac)$$

**TFI**: $\varphi: A \to S$ morf de inele $\Rightarrow A/\ker\varphi \overset{\simeq}{\hat{\varphi}} \operatorname{Im}\varphi$ izom de inele

$$\left(A/\ker\varphi \overset{\simeq}{\hat{\varphi}} \operatorname{Im}\varphi,\ \hat{\varphi}(\hat{\alpha}) = \varphi(\alpha)\right)$$

Aplicații:

1. $\mathbb{R}[X]/(X-a) \simeq \mathbb{R}$

Trebuie să construiesc un morf. de inele

$\varphi: \mathbb{R}[X] \to \mathbb{R}$ a.i.:

1. $\ker\varphi = (X-a)$ $\left(\Leftarrow (X-a)\in\ker\varphi,\ \varphi(X-a)=0,\ \varphi(X)=\varphi(a)\right)$
2. $\mathbb{R} = \operatorname{Im}(\varphi)$ ($\Leftrightarrow \varphi$ surjectiv)

Folosind proprietatea de universalitate a inelului de polin., definim morfismul de inele. (o)

$$\varphi(\underbrace{a_0+a_1X+\dots+a_nX^n}_{P(X)}) = a_0+a_1a+\dots+a_na^n$$

$$\varphi(P(X)) = P(a)$$

Fie $b\in\mathbb{R}$ $\varphi(b)=b \Rightarrow \varphi$ e surjectivă (◊)

(sau $\varphi(\underbrace{X+(b-a)}_{P(X)}) = b$ $\quad P(a)=a+b-a=b$)

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1. „$\supseteq$" (din construcție)

Fie $g(x) \in (x-a) \Rightarrow g(x) = (x-a)\cdot h(x)$, $h(x) \in \mathbb{R}[X]$

$\varphi(g(x)) = \varphi((x-a)\cdot h(x)) \overset{\varphi \text{ morf de inele}}{=} \varphi(x-a)\cdot\varphi(h(x)) =$

$= 0 \cdot h(a) = 0 \Rightarrow g(x) \in \ker\varphi$

Cum $g(x)$ a fost ales arbitrar $\Rightarrow (x-a) \subseteq \ker\varphi$

„$\subseteq$" (i.u T. Imp cu rest)

Fie $P(x) \in \ker\varphi \overset{\text{def}}{\Rightarrow} \varphi(P(x)) = 0 \overset{\text{def } \varphi}{\Rightarrow} P(a) = 0 \overset{\text{Bezout}}{\Rightarrow}$

$\Rightarrow P(x) = (x-a)\cdot g(x) + \underbrace{P(a)}_{0}$

$P(x) = (x-a)g(x)$, $g(x) \in \mathbb{R}[x] \Rightarrow P(x) \in (x-a) \Rightarrow$

$\ker\varphi \subseteq (x-a)$

Deci $\ker\varphi = (x-a)$ $(\square)$

Din $(\circ)$, $(\diamond)$, $(\square)$ $\overset{\text{TFI}}{\Rightarrow} \mathbb{R}[X]/(x-a) \simeq \mathbb{R}$

2. $\mathbb{Z}[x]/(x^2+1) \simeq \mathbb{Z}[i] = \{a+bi \mid a,b \in \mathbb{Z}\}$

$\{P(i) \mid P(x) \in \mathbb{Z}[x]\}$

Trebuie să construiesc un morfism de inele surjectiv:

$\varphi : \mathbb{Z}[x] \to \mathbb{Z}[i]$ a.î.

$\ker\varphi = (x^2+1)$

~~„$\supseteq$" (din construcție)~~

Fie $\varphi : \mathbb{Z}[x] \to \mathbb{Z}$ morfismul de inele (din prop. univ. a inelului de polinoame)

$\varphi(P(x)) = P(i)$

Fie $(a+bi) \in \mathbb{Z}[i] \Rightarrow \varphi(\underbrace{a+bx}_{P(x)}) = a+bi = \varphi$ e surj.

Vrem $\ker\varphi = (x^2+1)$

„$\supseteq$" (din construcție)

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Fie $g(x) \in (x^2+1) \Rightarrow g(x) = (x^2+1)\cdot h(x)$, $h(x) \in \mathbb{Z}[x]$.

$\varphi(g(x)) = \varphi((x^2+1)\cdot h(x)) \xrightarrow[\text{morf de inele}]{\varphi} \varphi(x^2+1)\cdot\varphi(h(x)) =$

$= 0 \cdot h(a) = 0 \Rightarrow g(x) \in \ker\varphi$ (1)

PT

Cum $g(x)$ a fost ales arbitrar $\Rightarrow (x^2+1) \subseteq \ker\varphi$

"$\subseteq$" (cu T. Cmp ce rest)

Fie $P(x) \in \ker\varphi \xrightarrow[\ker\varphi]{\text{def}} \varphi(P(x)) = 0 \xrightarrow[\varphi]{\text{def}} P(i) = 0 \xrightarrow{\text{Bezout}}$

$P(x) = (x^2+1)\cdot g(x) + r(x) \Rightarrow$ unde $\text{grad}(r(x)) < 2$, i.e. $ax+b$ cu $a,b \in \mathbb{Z}$

$\Rightarrow P(x) = (x^2+1)\cdot g(x) + \underbrace{P(i)}_{0}$

( $\hookrightarrow P(x) = (x^2+1)\cdot g(x)$, $g(x) \in \mathbb{R}[x] \Rightarrow P(x) \in (x^2+1) \Rightarrow$

$\Rightarrow \ker\varphi \subseteq (x^2+1) \Rightarrow \ker\varphi = (x^2+1)$ )

Cum $p(i)=0 \Rightarrow (i^2+1)g(i) + ai + b = 0 \Rightarrow ai + b = 0$

$a = b = 0 \Rightarrow r(x) = 0 \Rightarrow$

$\Rightarrow P(x) = (x^2+1)\cdot g(x) \Rightarrow P(x) \in (x^2+1) \Rightarrow$

$\Rightarrow \ker\varphi \subseteq (x^2+1) \Rightarrow \ker\varphi = (x^2+1)$ (2) $\xrightarrow{T\ 1\ i}$ $\mathbb{Z}[x]/(x^2+1) \simeq \mathbb{Z}[i]$

Dim 1 și 2 avem că $\ker\varphi = (x^2+1)$

TEMĂ $\rightarrow \mathbb{Z}[x]/(x^2-p_i) \simeq \mathbb{Z}[\sqrt{p_i}] = \{a+b\sqrt{p_i} \mid a,b \in \mathbb{Z}\}$

$i = \overline{1,17}$, $i = 3$, $p_3 = 5$

$\mathbb{Z}[x]/(x^2-5) \simeq \mathbb{Z}[\sqrt{5}] = \{a+b\sqrt{5} \mid a,b \in \mathbb{Z}\}$

Trebuie să construiesc un morfism de inele surjectiv:

$\varphi : \mathbb{Z}[x] \to \mathbb{Z}[\sqrt{5}]$ a.î. $\ker\varphi = (x^2-5)$.

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